

Interview Questions & Answers

SQL • Power BI • Data Modeling • ETL/DWH • Python/API • PySpark • Microsoft Fabric

Interview-ready answers with simple explanations and examples.

1. SQL Interview Answers

What is a JOIN?

Answer: A JOIN is used to combine data from two or more tables based on a related column.

```
SELECT c.customer_name, o.order_id
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id;
```

Remember: Hindi: Related data from two tables ko combine karne ke liye JOIN use karte hain.

INNER JOIN vs LEFT JOIN

Answer: INNER JOIN returns only the matching records from both tables, while LEFT JOIN returns all records from the left table and matching records from the right table.

Remember: Example: If there are 100 customers and 80 have orders, INNER JOIN returns 80 customers, while LEFT JOIN can return all 100.

WHERE vs HAVING

Answer: WHERE filters individual rows before grouping, whereas HAVING filters groups after GROUP BY.

```
SELECT city, COUNT(*) AS total_customers
FROM customers
WHERE status = 'Active'
GROUP BY city
HAVING COUNT(*) > 10;
```

Remember: WHERE → rows filter. HAVING → groups filter.

GROUP BY

Answer: GROUP BY is used to group rows having the same values so that we can perform aggregate calculations such as COUNT, SUM, AVG, MIN and MAX.

```
SELECT city, COUNT(*) AS customer_count
FROM customers
GROUP BY city;
```

Find duplicate records

Answer: Group by the column that should be unique and filter groups whose count is greater than one.

```
SELECT email, COUNT(*) AS cnt
FROM customers
GROUP BY email
HAVING COUNT(*) > 1;
```

Remember: Emails with count greater than 1 are duplicates.

What is ROW_NUMBER()?

Answer: ROW_NUMBER() is a window function that assigns a unique sequential number to each row within a result set or partition.

```
SELECT employee_id, salary,
       ROW_NUMBER() OVER (ORDER BY salary DESC) AS rn
FROM employees;
```

RANK vs DENSE_RANK

Answer: RANK gives the same rank to tied values and skips the next rank, while DENSE_RANK gives the same rank to tied values but does not skip ranks.

Example: Example: Salaries 10000, 9000, 9000, 8000 → RANK: 1,2,2,4; DENSE_RANK: 1,2,2,3.

Primary Key vs Foreign Key

Answer: A primary key uniquely identifies each record in a table. A foreign key is used to create a relationship between two tables by referencing a key from another table.

Example: Example: Customers.customer_id = Primary Key; Orders.customer_id = Foreign Key.

What is a Subquery?

Answer: A subquery is a query written inside another SQL query. It is used when the result of one query is required by another query.

```
SELECT *
FROM employees
WHERE salary > (
    SELECT AVG(salary)
    FROM employees
);
```

Find top 5 products by revenue

Answer: Group sales by product, calculate total revenue, sort in descending order, and return the top five.

```
SELECT product_id,
       SUM(revenue) AS total_revenue
FROM sales
GROUP BY product_id
ORDER BY total_revenue DESC
LIMIT 5;
```

What is NULL?

Answer: NULL represents a missing or unknown value in SQL. It is not equal to zero or an empty string.

```
SELECT *
FROM employees
WHERE manager_id IS NULL;
```

Remember: Use IS NULL or IS NOT NULL to check NULLs.

UNION vs UNION ALL

Answer: UNION combines results and removes duplicate rows, while UNION ALL combines results and keeps duplicates.

Remember: UNION → duplicates removed. UNION ALL → duplicates kept.

What is an Index?

Answer: An index is a database structure that helps SQL find rows faster, especially when searching or filtering on frequently used columns. Indexes require storage and can slow some INSERT or UPDATE operations.

Normalization vs Denormalization

Answer: Normalization organizes data into multiple related tables to reduce redundancy, while denormalization combines data to reduce joins and improve read performance.

Remember: Normalization → less duplicate data. Denormalization → faster/easier reads.

2. Power BI Interview Answers

What is Power BI used for?

Answer: Power BI is a business intelligence and data visualization tool used to connect to data sources, transform and model data, create interactive reports and dashboards, and share insights with users.

What steps do you follow to build a Power BI report?

Answer: First, connect to the data source. Then clean and transform data using Power Query. Next create the data model and relationships, create required DAX measures, build visuals, validate numbers, and publish the report to Power BI Service.

Dashboard vs Report

Answer: A report is usually a multi-page interactive analysis containing multiple visuals, while a dashboard is a single-page view used to monitor key metrics and KPIs.

What is a Semantic Model?

Answer: A semantic model is the data model used by Power BI to organize tables, relationships, measures, and business logic so users can analyze data easily.

What is Power Query?

Answer: Power Query is used to connect, extract, clean, transform, and prepare data before loading it into the Power BI model.

Remember: Examples: remove duplicates, handle nulls, change data types, merge and append tables.

Measure vs Calculated Column

Answer: A calculated column is calculated row by row and stored in the model, while a measure is calculated dynamically based on filter context when it is used in a visual.

```
Total Sales = SUM(Sales[Amount])
```

Remember: Column → row-level calculation. Measure → dynamic aggregation.

Import vs DirectQuery

Answer: In Import mode, data is loaded into Power BI's in-memory model, which generally provides fast report performance. In DirectQuery, Power BI queries the underlying data source when the report needs data.

What is a Slicer?

Answer: A slicer is a visual filter that allows users to interactively filter the data displayed in other visuals.

Remember: Example: Year slicer filters charts and tables by selected year.

What is Refresh?

Answer: Refresh means updating the Power BI model with the latest data from the source.

Remember: Refresh can fail because of source connection problems, changed credentials, renamed columns/tables, data type issues, gateway problems, or an unavailable source.

SUM vs SUMX

Answer: SUM adds values from a single column, while SUMX iterates through a table row by row, evaluates an expression for each row, and then adds the results.

```
Total Sales = SUM(Sales[Amount])
```

```
Total Sales =  
SUMX(  
    Sales,  
    Sales[Quantity] * Sales[Price]  
)
```

Remember: SUM → existing column total. SUMX → row-by-row calculation followed by total.

SUM vs CALCULATE

Answer: SUM performs a simple aggregation, while CALCULATE evaluates an expression under a modified filter context.

```
Total Sales = SUM(Sales[Amount])
```

```
USA Sales =  
CALCULATE(  
    SUM(Sales[Amount]),  
    Sales[Country] = "USA"  
)
```

Row Context vs Filter Context

Answer: Row context means evaluating a calculation for the current row, while filter context is the set of filters that determine which data is included in a calculation.

Remember: Row Context → one current row. Filter Context → which data is included.

What is Context Transition?

Answer: Context transition happens when CALCULATE converts the current row context into filter context.

YTD Sales

Answer: YTD Sales calculates sales from the beginning of the year up to the current date.

```
YTD Sales =  
TOTALYTD(  
    [Total Sales],  
    'Date'[Date]  
)
```

MoM Growth %

Answer: MoM Growth measures the percentage change in sales compared with the previous month.

```
MoM Growth % =  
DIVIDE(  
    [Current Month Sales] - [Previous Month Sales],  
    [Previous Month Sales]  
)
```

How do you debug wrong numbers in DAX?

Answer: First, check source data and relationships. Then verify filter context and data types. Break the measure into smaller measures, test intermediate results in a table visual, and check whether the issue is relationships, filters, or DAX logic.

Remember: This is an important practical interview answer.

3. Data Modeling

What is a Fact Table?

Answer: A fact table contains measurable business transactions or events, such as sales, quantity, revenue, or orders.

What is a Dimension Table?

Answer: A dimension table contains descriptive information used to analyze facts, such as customer, product, date, or location.

Why Star Schema?

Answer: Star schema is preferred because it is simple, easy to understand, provides efficient filtering and aggregation, and generally works well with Power BI's analytical model.

What is a Date Table?

Answer: A date table is a dedicated table containing dates and related attributes such as year, quarter, month, and day. It is useful for time-intelligence calculations such as YTD, MTD, and YoY.

What is a Surrogate Key?

Answer: A surrogate key is a system-generated unique key used to identify records in a dimension table. It is usually numeric and independent of the business key.

4. ETL + Data Warehouse

What is ETL?

Answer: ETL stands for Extract, Transform, Load. We extract data from source systems, transform and clean it according to business requirements, and then load it into a target system such as a data warehouse.

Full Load vs Incremental Load

Answer: Full load loads all data every time, while incremental load loads only new or changed data.

Remember: Full → everything. Incremental → only changes.

What is Data Quality?

Answer: Data quality means ensuring that data is accurate, complete, consistent, valid, unique, and timely enough for its intended use.

Remember: Checks: NULLs, duplicates, data types, ranges, referential integrity, uniqueness, completeness.

What is SCD?

Answer: SCD stands for Slowly Changing Dimension. It is a technique used to manage changes in dimension data over time.

SCD Type 1 vs Type 2

Answer: SCD Type 1 overwrites the old value and does not maintain history. SCD Type 2 keeps historical values by creating a new record when the dimension changes.

Remember: Type 1 → old value lost. Type 2 → historical and current records maintained.

5. Python / Pandas / API

How do you read a CSV in Python?

Answer: Use pandas' read_csv function to load a CSV file into a DataFrame.

```
import pandas as pd
df = pd.read_csv("sales.csv")
```

How do you handle missing values?

Answer: First identify missing values. Depending on the business requirement, remove them, fill them with an appropriate value, or investigate the source.

```
df.isnull().sum()
df.dropna()
df["salary"] = df["salary"].fillna(0)
```

List vs Dictionary

Answer: A list stores an ordered collection of values and is accessed using indexes, while a dictionary stores data as key-value pairs and is accessed using keys.

```
my_list = [10, 20, 30]
my_dict = {"name": "John", "salary": 50000}
```

What is groupby in Pandas?

Answer: Pandas groupby is used to group data based on one or more columns and perform aggregation operations such as sum, count, or average.

```
df.groupby("city")["sales"].sum()
```

What is an API?

Answer: API stands for Application Programming Interface. It allows different applications or systems to communicate and exchange data.

GET vs POST

Answer: GET is generally used to retrieve data, while POST is generally used to send data to the server to create a new resource or perform an operation.

Pandas vs PySpark

Answer: Pandas is suitable for smaller datasets that can generally fit into memory on one machine, while PySpark is designed for large-scale data processing across distributed systems.

Transformation vs Action in Spark

Answer: A transformation creates a new DataFrame or RDD and is evaluated lazily, while an action triggers actual execution and returns a result or writes data.

```
# Transformation
df.filter(df["salary"] > 50000)

# Action
df.count()
```

6. Microsoft Fabric

What is Microsoft Fabric?

Answer: Microsoft Fabric is an end-to-end analytics platform from Microsoft that brings together data integration, data engineering, data warehousing, data science, real-time analytics, and Power BI in one platform.

Fabric vs only Power BI

Answer: Power BI mainly focuses on data modeling, visualization, reporting, and business intelligence. Microsoft Fabric provides a broader end-to-end data and analytics platform including data integration, engineering, lakehouse, data warehouse, real-time analytics, and Power BI.

What is OneLake?

Answer: OneLake is the centralized data lake for Microsoft Fabric. It provides a single logical place to store and access organizational data across different Fabric workloads.

Lakehouse vs Data Warehouse in Fabric

Answer: A Lakehouse is suitable when we need to work with large volumes of structured and unstructured data and support data engineering or data science workloads. A Data Warehouse is more suitable for structured relational data and traditional SQL-based analytics and reporting.

What is a Fabric Pipeline?

Answer: A Fabric pipeline is used to orchestrate and automate data movement and transformation workflows. It can connect to different data sources, move data, and execute data processing activities according to a defined schedule or workflow.

7. Project / Communication

Walk me through your Power BI project.

Answer: In my project, I worked on a sales dashboard. First, I collected data from the source systems. I used Power Query to clean and transform the data, handled missing values and duplicates, and then created a star-schema data model with fact and dimension tables. After that, I created DAX measures for KPIs such as total sales, YTD sales and growth percentage. I then created interactive visuals using charts, cards and slicers. I validated the numbers against the source data and finally published the report to Power BI Service and configured refresh.

How do you handle wrong numbers in a report?

Answer: I first compare the report numbers with the source data. Then I check relationships, filters, data types, duplicate records, and DAX logic. I isolate the issue by testing smaller calculations and intermediate results before fixing and revalidating the report.

What challenges did you face in your project and how did you fix them?

Answer: A good structure is: explain the challenge, explain how you investigated it, describe the solution, and mention how you validated the result. For example, if numbers did not match the source, I checked relationships, duplicate records, filters, and DAX logic before correcting the model or measure.